NOTE: the provisions of Chapter 4A of Orange Development Control Plan 2004 will be effective upon gazettal of Amendment 24 to Orange Local Environmental Plan 2011

Contents

4A FLOOD AFFECTED LAND 4A.1
4A.1 INTRODUCTION 4A.1
4A.1.1 What does the plan do? 4A.1
4A.1.2 Objectives 4A.2
4A.1.3 Will the plan affect my property? 4A.2
4A.2 HOW TO USE THIS PLAN 4A.2
4A.3 WHAT ARE THE CRITERIA FOR DETERMINING APPLICATIONS? 4A.3
4A.3.1 General 4A.3
4A.3.2 Land use categories 4A.3
4A.3.3 What controls apply to proposed residential developments? 4A.3
4A.3.4 What if my allotment is partially in the Floodway or Overland Flow? 4A.3
4A.3.5 Fencing 4A.4
4A.3.6 Other Uses and works 4A.4
4A.3.7 Other documents which may need to be read in conjunction with this plan: 4A.4
4A.3.8 What Information Do You Have to Submit to Council? 4A.4
4A.4 DESCRIPTION OF TERMS 4A.5
4A.5 ANNEXURE 1 - INTERIM FLOOD PLANNING AREAS AND LEVELS 4A.7
4A.6 ANNEXURE 2 - LAND USE CATEGORIES 4A.15
4A.6.1 Essential Community Facilities 4A.15
4A.6.2 Critical Utilities 4A.15
4A.6.3 Subdivision and Filling 4A.15
4A.6.4 Residential 4A.15
4A.6.5 Commercial or Industrial 4A.15
4A.6.6 Recreation or Agricultural 4A.16
4A.6.7 Minor Development 4A.16
4A.7 ANNEXURE 3 - PRESCRIPTIVE DEVELOPMENT CONTROLS – FLOOD RESPONSE LEVELS 4A.17
4A.8 ANNEXURE 4 - FLOOD COMPATIBLE MATERIALS 4A.19

4A FLOOD AFFECTED LAND

4A.1 INTRODUCTION

This chapter of the Development Control Plan (DCP) was prepared to provide specific development controls to guide development of flood affected land within Orange.

The DCP incorporates the findings of the Blackmans Swamp and Ploughmans Creek Flood Study and the procedures set out in the NSW Floodplain Management Manual, 2005.

4A.1.1 What does the plan do?

The plan provides information and guidelines to assist people considering the development or use of land affected by potential flooding within Orange. Development may include, among other things:

- building construction;
- alterations and additions to existing developments;
- filling land; earthworks including but not limited to: filling, excavation, trenching , swale drains,

- landscape such as; garden edging, planting of trees and shrubs, construction of pathways and other hard surface areas, construction of fences;
- establishing a business;
- subdividing land;
- minor development such as Carports and Garages; and
- permit merit based assessment of development upon an allotment where it can be demonstrated that the development site is above the 1% AEP (100 year ARI).

4A.1.2 Objectives

This chapter of the DCP aims to:

a) Minimise the potential impact of development and other activity upon the aesthetic, recreational and ecological value of the waterway corridors.

b) Increase public awareness of the hazard and extent of land affected by all potential floods, including floods greater than the Adopted Flood and to ensure essential services and land uses are planned in recognition of all potential floods.

c) Inform the community of Council's policy for the use and development of flood prone land.

d) Reduce the risk to human life and damage to property caused by flooding through controlling development on land affected by potential floods.

e) Provide detailed controls for the assessment of applications lodged in accordance with the Environmental Planning and Assessment Act 1979 on land affected by potential floods.

f) Provide different guidelines, for the use and development of land subject to all potential floods in the floodplain, which reflect the probability of the flood occurring and the potential hazard within different areas.

g) Apply a "merit-based approach" to all development decisions which takes account of social, economic and ecological as well as flooding considerations.

h) Control development and other activity within each of the individual floodplains within the LGA having regard to the characteristics and level of information available for each of the floodplains.

i) Deal equitably and consistently with applications for development on land affected by potential floods, in accordance with the principles contained in the NSW Floodplain Development Manual, 2005 issued by the NSW Government.

j) Provide controls for the assessment of applications on land affected by potential floods in accordance with the provisions of Orange LEP 2011 (as amended).

4A.1.3 Will the plan affect my property?

The DCP applies to all development permissible with the consent of Council on land in Orange that is shown in Annexure 1.0.

4A.2 HOW TO USE THIS PLAN

The DCP provides a general criterion for determining development applications. The criteria recognise that different controls apply to different land uses and levels of potential flood inundation or hazard.

Step 1 Using the maps in Annexure 1 identify whether your land is flood affected land, and whether it is within the Floodway (Main Stream Flooding) (red areas) or if it is subject to Overland Flow Deeper than 100mm (blue areas).

Step 2 Using Annexure 2 identify the most relevant land use category for your proposed development or works.

Step 3 Refer to the matrix in Annexure 3 to determine the Flood Response Levels for different aspects of your proposed development or works – such as floor levels, building components, structural soundness, evacuation, flood affectation and management and design.

Step 4 Using the Flood Response Levels from step 3 consult the Flood Response Controls in Annexure 3 and incorporate the requirements into your design.

Step 5 Where flood compatible materials are required by a Flood Response Control consult Annexure 4 for guidance.

4A.3 WHAT ARE THE CRITERIA FOR DETERMINING APPLICATIONS?

4A.3.1 General

The criteria for determining applications for proposals potentially affected by flooding are structured in recognition that different controls are applicable to different land uses and levels of potential flood inundation and hazard.

4A.3.2 Land use categories

Seven land use categories have been adopted and are listed in Annexure 2.0 - ALL development shall be deemed to be within one of the 7 categories. Where a development is of a type not explicitly listed, it shall be deemed to be within the category that contains the most similar forms of development.

4A.3.3 What controls apply to proposed residential developments?

The development controls apply to all flood prone land at or below the 1% AEP flood. The type of control has been graded relative to the severity and frequency of potential floods, having regard to the two following applicable categories:

- *Floodway – Shown in red on the Interim Flood Planning Areas and Levels Main Stream Flooding and Overland Flow affected Areas Maps shown in Annexure 1*
- *Overland Flow – Shown in blue on the Interim Flood Planning Areas and Levels Main Stream Flooding and Overland Flow affected Areas Maps shown in Annexure 1*
- *Annexure 3 outlines the flood response levels and controls relevant to the area to which this Policy applies for each of the above categories*

4A.3.4 What if my allotment is partially in the Floodway or Overland Flow?

Any proposed development on that land shall be subject to a merit based assessment with respect to the proposed development's location and impact on flood affectation.

4A.3.5 Fencing

Any proposed fencing is to be shown on the plans accompanying a development application and the likely effect of such fencing on flood behaviour is to be assessed by a suitably qualified engineer.

All fencing panels (including Colorbond, galvanised, timber, wire and brush) to be either:

a) removable so that panels can be laid flat; or

b) horizontally hinged where a portion of at least 1.2m high is capable of:

   i) Swinging open to allow floodwater to pass.
   ii) Raised as to allow floodwaters to flow beneath

Fencing may be required to comply with a type and siting criteria as prescribed by Council.

4A.3.6 Other Uses and works

All other development, building or other works within any of the categories that require Council's consent will be considered on their merit. In consideration of such applications Council must determine that the proposed development is in compliance with the objectives of this policy.

4A.3.7 Other documents which may need to be read in conjunction with this plan:

- Orange Local Environmental Plan 2011;
- Relevant Council policies, development control plans and specifications;
- Blackmans Swamp and Ploughman's Creek Flood Study;
- NSW Government Floodplain Management Manual;
- Environmental Protection Authority Guidelines on Sediment and Erosion Control;
- Environmental Protection Authority Guidelines on on-site effluent disposal.

4A.3.8 What Information Do You Have to Submit to Council?

In order to obtain the approval of Council you must lodge a Development Application in accordance with the Environmental Planning and Assessment Act 1979.

The application must be accompanied by:

- a site plan showing the location of proposed works or structures;
- contour lines of the site to enable assessment of likely flood behaviour;
- materials proposed to be used in the development;
- proposed finished floor levels; and

- the location and proposed height of all electrical and critical utilities.

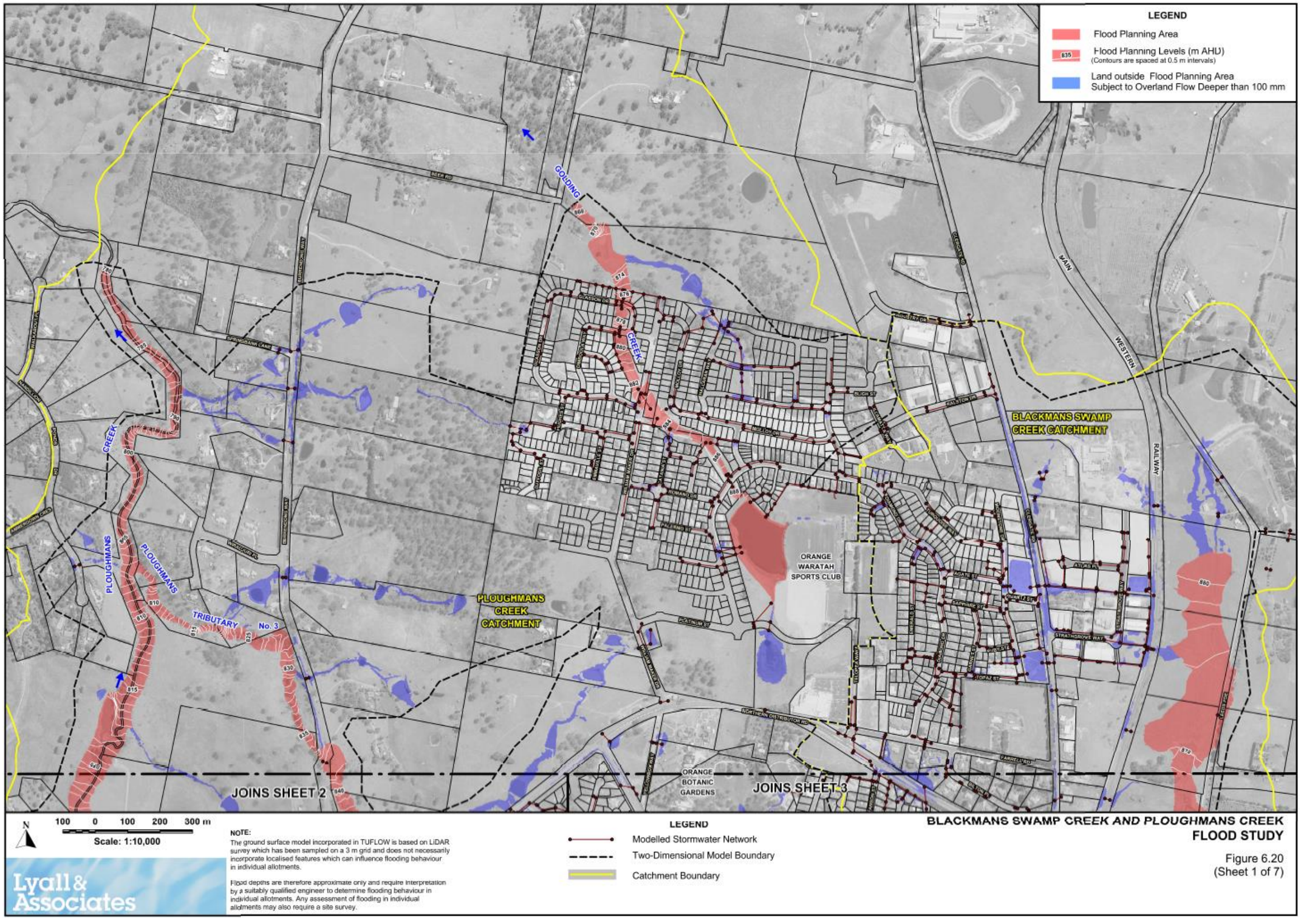
4A.4 DESCRIPTION OF TERMS

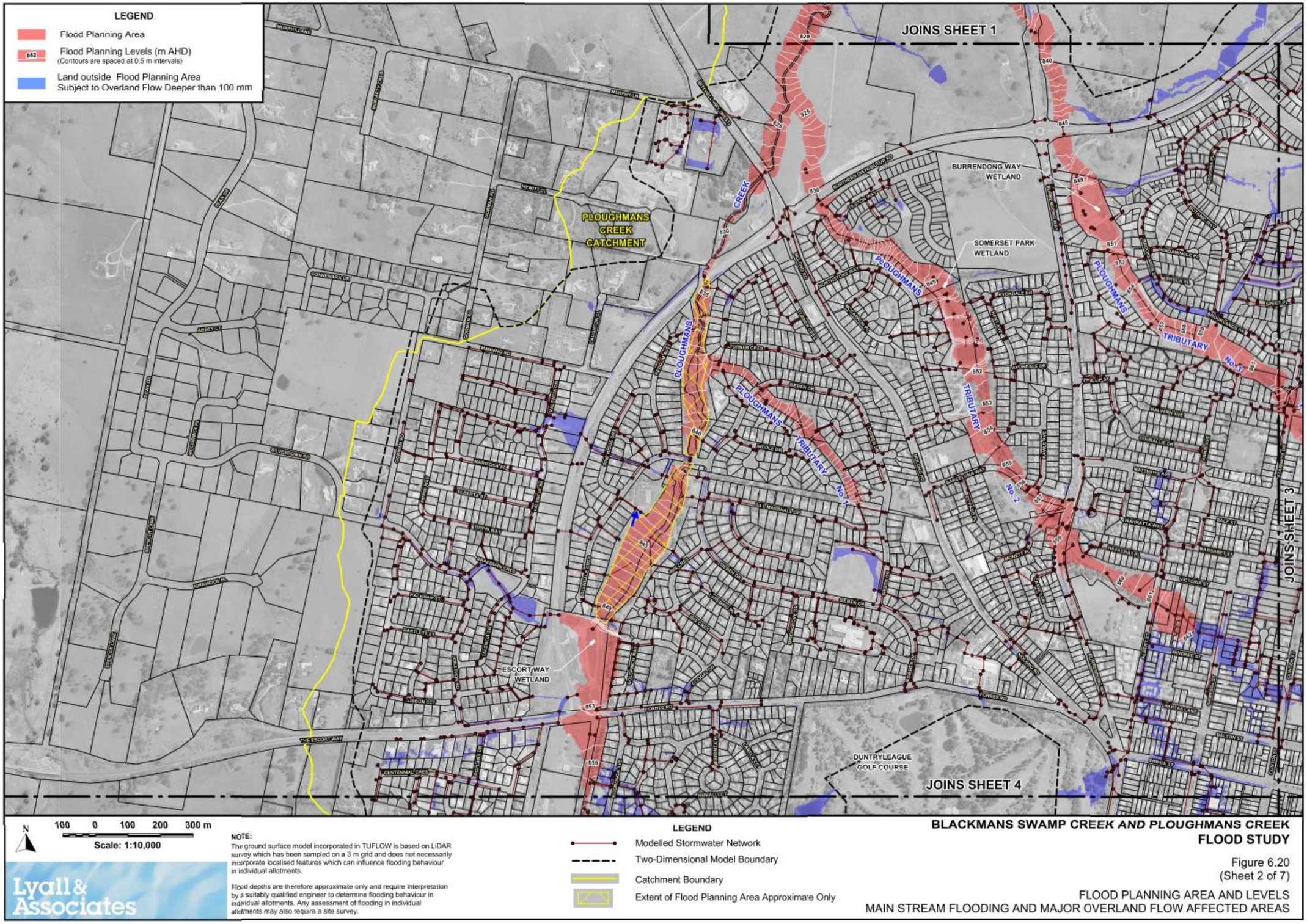
Definitions of terms used in the DCP are provided as follows:

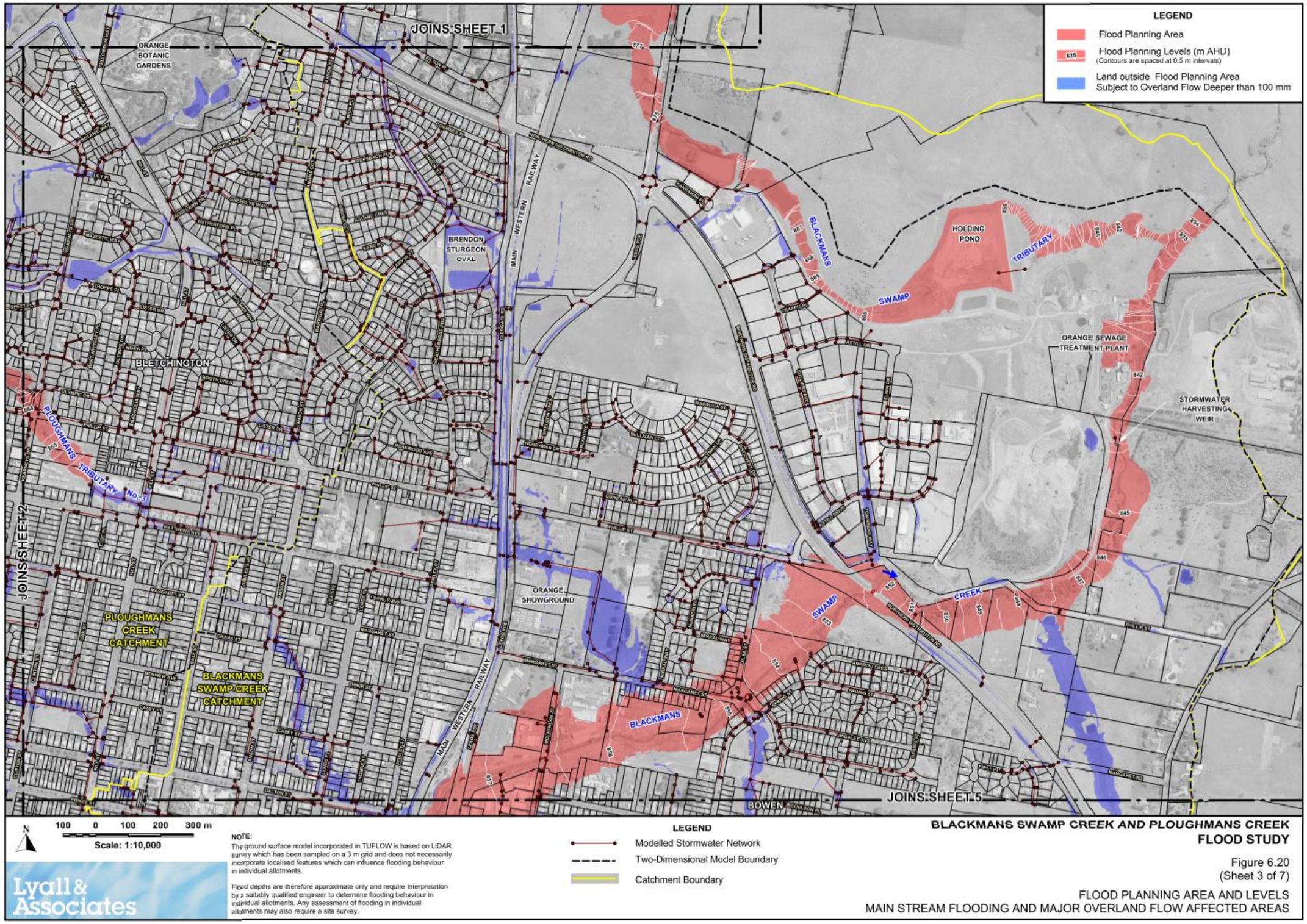
Term	Description
Australia Height Datum (AHD)	A common National plain of level corresponding approximately to mean sea level.
Annual exceedance probability (AEP)	The probability that a given rainfall total accumulated over a given duration will be exceeded in any one given year. (Source: Bureau of Meteorology).
Design floor level	The floor level specified in this policy which applies to the relevant land use type and the location and existing ground level of the site.
Designated flood	The flood adopted for planning purposes relative to the sensitivity of different land uses and the flood risk, as specified by *Annexure 1.0.*
Designated flood level	Represents the maximum level of water reached measured as a height above Australian Height Datum during a designated flood.
Flood awareness	An appreciation of the likely effects of flooding and knowledge of the relevant flood warning and evacuation procedures.
Flood compatible building components	A combination of measures incorporated in the design and/or construction and alteration of individual buildings or structures subject to flooding, and the use of flood compatible materials for the reduction or elimination of flood damage as indicated in the Floodplain Development Manual.
Flood compatible materials	Those materials used in building which are resistant to damage when inundated. A list of flood compatible materials is attached in Annexure 4.0.
Flood evacuation strategy	The proposed strategy for the evacuation of areas during periods of flood as specified within any policy of Council, the Floodplain Risk Management Plan, by advice from the NSW State Emergency Service (SES) or as determined in the assessment of individual proposals.
Flood liable land	The area of land which is subject to inundation by floods up to and including the 1% AEP.
Flood planning level	A level as defined in a flood management study for the purposes of setting flood related control applicable to a particular use of the land.
Flood planning area	The area shown in red on the Flood Planning Maps within Annexure 1.
Floodplain	An area of land that is subject to inundation by the probable maximum flood or an extreme flood event.
Floodway	An area of land as defined in a flood study which constitutes the main flood path of floodwaters and in which flood hazard conditions occur, as distinctly identified and shown in red on Interim Flood Planning Areas.
Flood hazard	The potential risk to life and potential damage caused by flooding.
Flood prone land	Land susceptible to inundation by the maximum flood PMF or an appropriate extreme flood event.
Freeboard	A factor of safety usually expressed as a height above the designated flood. Freeboard tends to compensate for factors such as wave action, localised hydraulic effects and the like.
Habitable floor area	Refers to a room (other than a garage, shed, bathroom, laundry, WC or the like) that is constructed or adapted for domestic living such as a lounge room, dining room, rumpus room, kitchen, bedroom.
Interim designated flood event	Flood event based on historical information which allows the provisions of an Environmental Planning Instrument to be implemented until such time as a floodplain risk management plan has been adopted for an area,

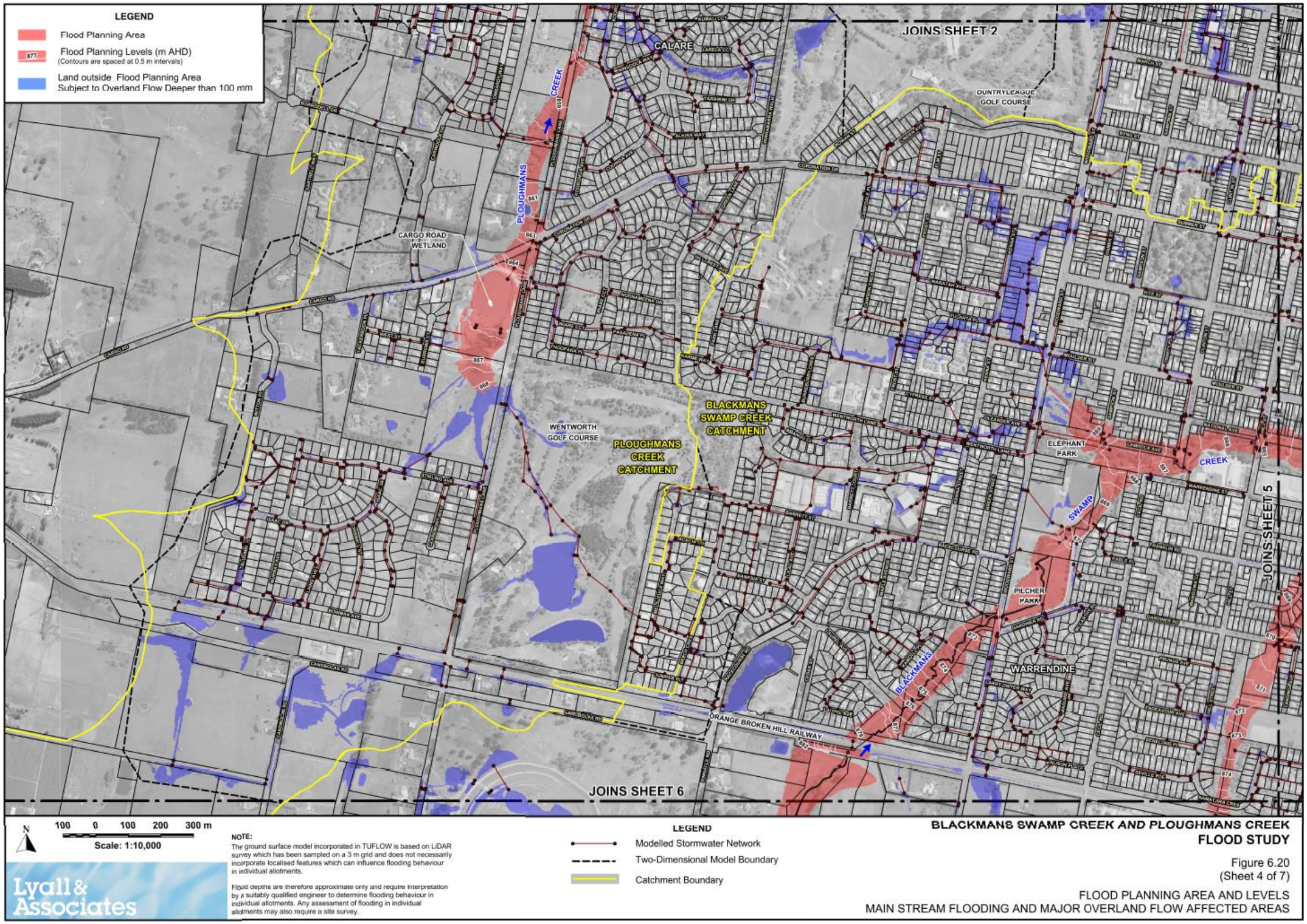
Term	Description
	the Interim Designated Flood Event determines the extent of flood liable land in the interim period.
Overland flow	An area of land as defined in a flood study which constitutes secondary flow path of floodwaters and in which flood hazard conditions occur, as distinctly identified and shown in blue on Interim Flood Planning Areas and Levels Maps within Annexure 1.
Probable maximum flood (PMF)	The largest flood that could conceivably occur at a particular location.
Reliable access during a flood	The ability for people to safely evacuate an area subject to imminent flooding within effective warning time and without a need to travel through areas where water depths increase.

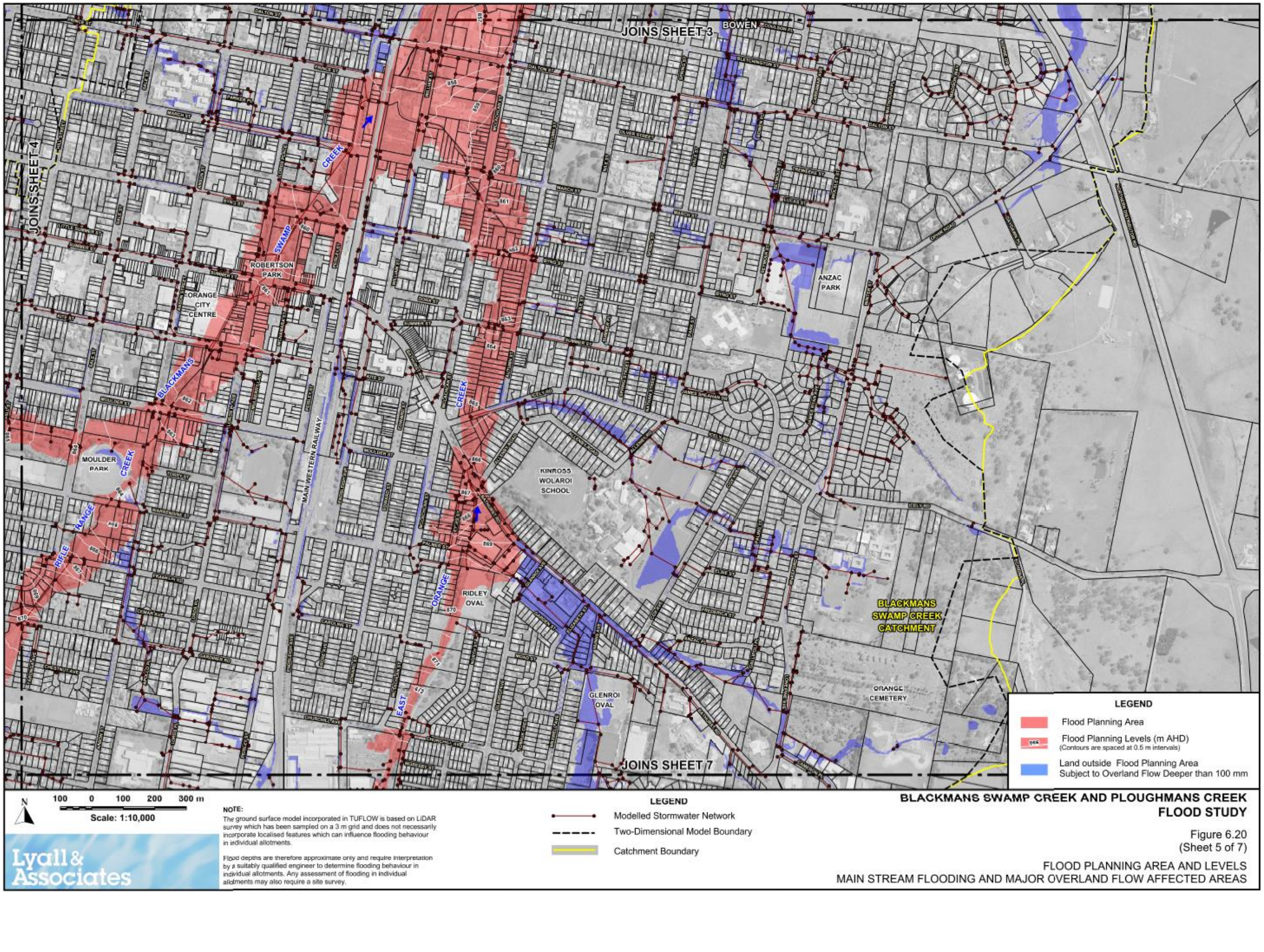
4A.5 ANNEXURE 1 - INTERIM FLOOD PLANNING AREAS AND LEVELS

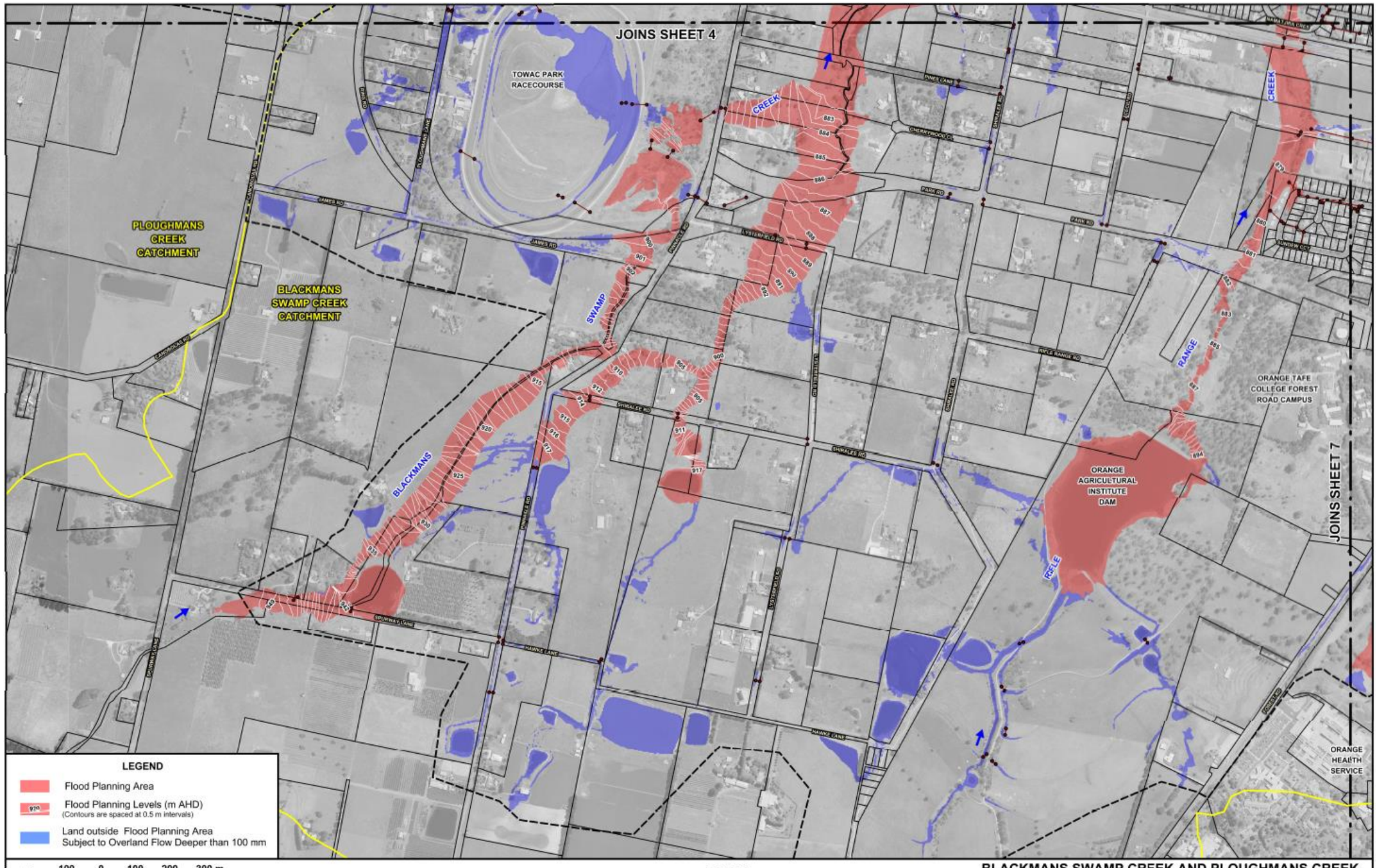

LEGEND
Flood Planning Area
835
Flood Planning Levels (m AHD)
(Contours are spaced at 0.5 m intervals)
Land outside Flood Planning Area
Subject to Overland Flow Deeper than 100 mm
GOLDING CREEK
CREEK
PLOUGHMANS CREEK
TRIBUTARY No. 3
BLACKMANS SWAMP CREEK CATCHMENT
PLOUGHMANS CREEK CATCHMENT
ORANGE WARATAH SPORTS CLUB
ORANGE BOTANIC GARDENS
JOINS SHEET 2
JOINS SHEET 3
MAIN WESTERN RAILWAY
BASER RD
SPRINGSIDE LANE
DIVISION RD
MACQUARIE AVE
ROYAL AVE
PALMERS LA
PLATINUM ST
AGATE ST
SAPPHIRE ST
TOPAZ ST
STRATHGROVE WAY
ATLAS PL
CAMBOON RD
100 0 100 200 300 m
Scale: 1:10,000
NOTE:
The ground surface model incorporated in TUFLOW is based on LiDAR survey which has been sampled on a 3 m grid and does not necessarily incorporate localised features which can influence flooding behaviour in individual allotments.
Flood depths are therefore approximate only and require interpretation by a suitably qualified engineer to determine flooding behaviour in individual allotments. Any assessment of flooding in individual allotments may also require a site survey.
LEGEND
Modelled Stormwater Network
Two-Dimensional Model Boundary
Catchment Boundary
Lyall & Associates
BLACKMANS SWAMP CREEK AND PLOUGHMANS CREEK FLOOD STUDY
Figure 6.20
(Sheet 1 of 7)

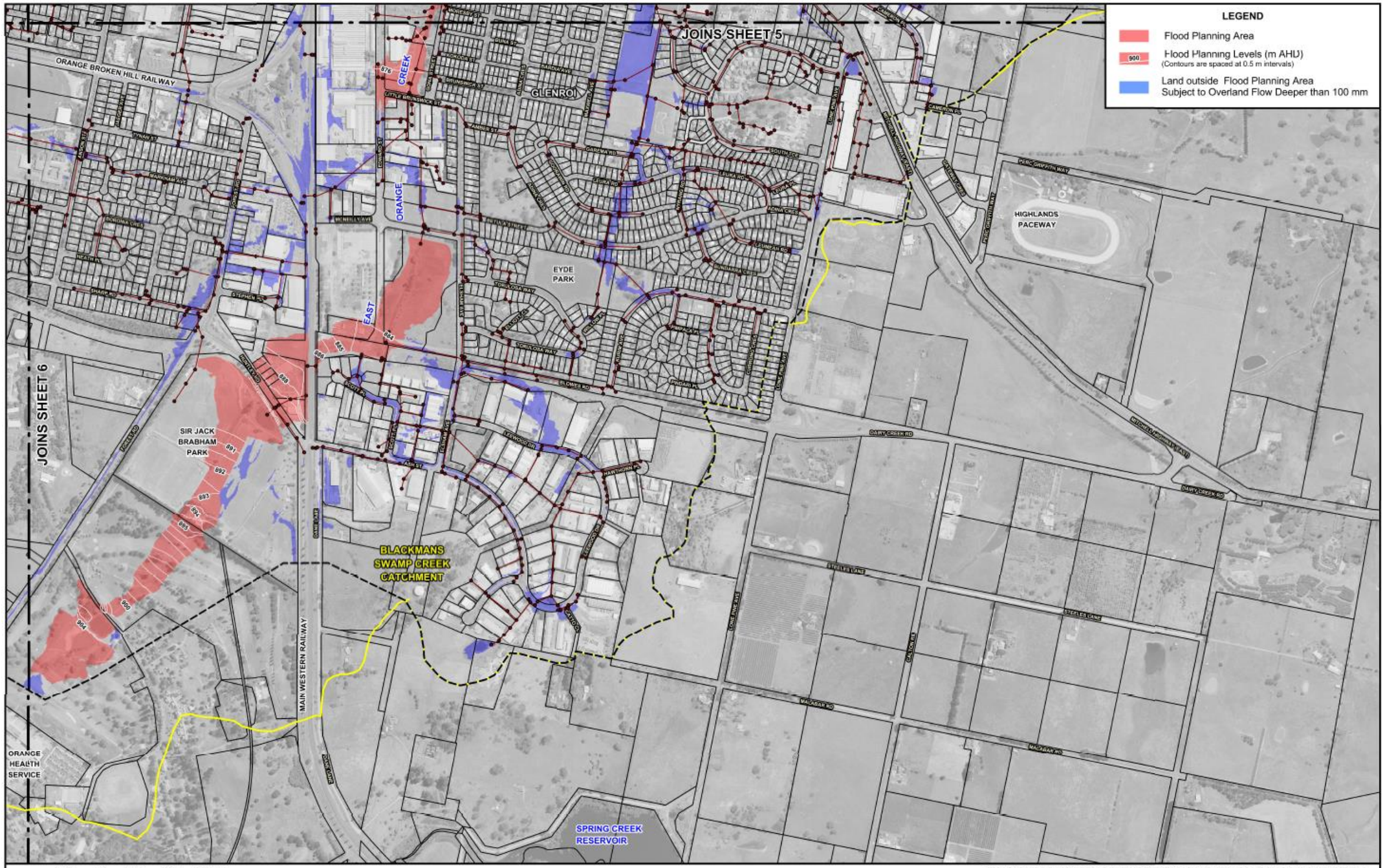

LEGEND
Flood Planning Area
852 Flood Planning Levels (m AHD)
(Contours are spaced at 0.5 m intervals)
Land outside Flood Planning Area
Subject to Overland Flow Deeper than 100 mm
JOINS SHEET 1
PLOUGHMANS CREEK CATCHMENT
BURRENDONG WAY WETLAND
SOMERSET PARK WETLAND
ESCORT WAY WETLAND
DUNTRYLEAGUE GOLF COURSE
JOINS SHEET 3
JOINS SHEET 4
PLOUGHMANS CREEK
TRIBUTARY No. 1
TRIBUTARY No. 2
TRIBUTARY No. 3
100 0 100 200 300 m
Scale: 1:10,000
NOTE:
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LEGEND
Modelled Stormwater Network
Two-Dimensional Model Boundary
Catchment Boundary
Extent of Flood Planning Area Approximate Only
Lyall & Associates
BLACKMANS SWAMP CREEK AND PLOUGHMANS CREEK FLOOD STUDY
Figure 6.20
(Sheet 2 of 7)
FLOOD PLANNING AREA AND LEVELS
MAIN STREAM FLOODING AND MAJOR OVERLAND FLOW AFFECTED AREAS


LEGEND
Flood Planning Area
Flood Planning Levels (m AHD)
(Contours are spaced at 0.5 m intervals)
Land outside Flood Planning Area
Subject to Overland Flow Deeper than 100 mm
JOINS SHEET 1
JOINS SHEET 2
JOINS SHEET 5
ORANGE BOTANIC GARDENS
BRENDON STURGEON OVAL
MAIN WESTERN RAILWAY
BLETCHINGTON
PLOUGHMANS TRIBUTARY No. 3
PLOUGHMANS CREEK CATCHMENT
BLACKMANS SWAMP CREEK CATCHMENT
BLACKMANS SWAMP
HOLDING POND
TRIBUTARY
ORANGE SEWAGE TREATMENT PLANT
STORMWATER HARVESTING WEIR
ORANGE SHOWGROUND
CREEK
BOWEN
N
100 0 100 200 300 m
Scale: 1:10,000
Lyall & Associates
NOTE:
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Flood depths are therefore approximate only and require interpretation by a suitably qualified engineer to determine flooding behaviour in individual allotments. Any assessment of flooding in individual allotments may also require a site survey.
LEGEND
Modelled Stormwater Network
Two-Dimensional Model Boundary
Catchment Boundary
BLACKMANS SWAMP CREEK AND PLOUGHMANS CREEK
FLOOD STUDY
Figure 6.20
(Sheet 3 of 7)
FLOOD PLANNING AREA AND LEVELS
MAIN STREAM FLOODING AND MAJOR OVERLAND FLOW AFFECTED AREAS


LEGEND
Flood Planning Area
Flood Planning Levels (m AHD)
(Contours are spaced at 0.5 m intervals)
Land outside Flood Planning Area
Subject to Overland Flow Deeper than 100 mm
JOINS SHEET 2
CALARE
COUNTRY LEAGUE GOLF COURSE
PLOUGHMANS CREEK
CARGO ROAD WETLAND
WENTWORTH GOLF COURSE
PLOUGHMANS CREEK CATCHMENT
BLACKMANS SWAMP CREEK CATCHMENT
ELEPHANT PARK
CREEK
SWAMP
PILCHER PARK
WARRENDINE
BLACKMANS
ORANGE BROKEN HILL RAILWAY
JOINS SHEET 5
JOINS SHEET 6
CARGO RD
CANOBOLAS RD
CORONATION DR
N
100 0 100 200 300 m
Scale: 1:10,000
NOTE:
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Flood depths are therefore approximate only and require interpretation by a suitably qualified engineer to determine flooding behaviour in individual allotments. Any assessment of flooding in individual allotments may also require a site survey.
LEGEND
Modelled Stormwater Network
Two-Dimensional Model Boundary
Catchment Boundary
Lyall & Associates
BLACKMANS SWAMP CREEK AND PLOUGHMANS CREEK FLOOD STUDY
Figure 6.20
(Sheet 4 of 7)
FLOOD PLANNING AREA AND LEVELS
MAIN STREAM FLOODING AND MAJOR OVERLAND FLOW AFFECTED AREAS


JOINS SHEET 3
JOINS SHEET 4
JOINS SHEET 7
BOWEN
ROBERTSON PARK
ORANGE CITY CENTRE
ANZAC PARK
MOULDER PARK
KINROSS WOLAROI SCHOOL
RIDLEY OVAL
GLENROI OVAL
ORANGE CEMETERY
BLACKMANS SWAMP CREEK CATCHMENT
SWAMP CREEK
BLACKMANS CREEK
MAIN WESTERN RAILWAY
RIFLE RANGE CREEK
ORANGE EAST
LEGEND
Flood Planning Area
Flood Planning Levels (m AHD)
(Contours are spaced at 0.5 m intervals)
Land outside Flood Planning Area
Subject to Overland Flow Deeper than 100 mm
N
100 0 100 200 300 m
Scale: 1:10,000
NOTE:
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LEGEND
Modelled Stormwater Network
Two-Dimensional Model Boundary
Catchment Boundary
Lyall & Associates
BLACKMANS SWAMP CREEK AND PLOUGHMANS CREEK FLOOD STUDY
Figure 6.20
(Sheet 5 of 7)
FLOOD PLANNING AREA AND LEVELS
MAIN STREAM FLOODING AND MAJOR OVERLAND FLOW AFFECTED AREAS


JOINS SHEET 4
TOWAC PARK RACECOURSE
PLOUGHMANS CREEK CATCHMENT
BLACKMANS SWAMP CREEK CATCHMENT
CREEK
SWAMP
BLACKMANS
RANGE
ORANGE TAFE COLLEGE FOREST ROAD CAMPUS
ORANGE AGRICULTURAL INSTITUTE DAM
ORANGE HEALTH SERVICE
JOINS SHEET 7
JAMES RD
PARK RD
LYSTER RIDGE RD
SHIRALEE RD
HAMKE LANE
SPRINGVALE LANE
RIFLE RANGE RD
LEGEND
Flood Planning Area
Flood Planning Levels (m AHD)
(Contours are spaced at 0.5 m intervals)
Land outside Flood Planning Area
Subject to Overland Flow Deeper than 100 mm
N
100 0 100 200 300 m
Scale: 1:10,000
Lyall & Associates
NOTE:
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Flood depths are therefore approximate only and require interpretation by a suitably qualified engineer to determine flooding behaviour in individual allotments. Any assessment of flooding in individual allotments may also require a site survey.
LEGEND
Modelled Stormwater Network
Two-Dimensional Model Boundary
Catchment Boundary
BLACKMANS SWAMP CREEK AND PLOUGHMANS CREEK FLOOD STUDY
Figure 6.20
(Sheet 6 of 7)
FLOOD PLANNING AREA AND LEVELS
MAIN STREAM FLOODING AND MAJOR OVERLAND FLOW AFFECTED AREAS


LEGEND
Flood Planning Area
Flood Planning Levels (m AHD)
(Contours are spaced at 0.5 m intervals)
Land outside Flood Planning Area
Subject to Overland Flow Deeper than 100 mm
JOINS SHEET 5
JOINS SHEET 6
ORANGE BROKEN HILL RAILWAY
GLENROI
EYDE PARK
HIGHLANDS PACEWAY
SIR JACK BRABHAM PARK
BLACKMANS SWAMP CREEK CATCHMENT
MAIN WESTERN RAILWAY
ORANGE HEALTH SERVICE
SPRING CREEK RESERVOIR
ORANGE CREEK
EAST ORANGE CREEK
BLOOMS RD
DARBY CREEK RD
MACQUARIE HIGHWAY
WALASAR RD
STEELS LANE
N
100 0 100 200 300 m
Scale: 1:10,000
NOTE:
The ground surface model incorporated in TUFLOW is based on LiDAR survey which has been sampled on a 3 m grid and does not necessarily incorporate localised features which can influence flooding behaviour in individual allotments.
Flood depths are therefore approximate only and require interpretation by a suitably qualified engineer to determine flooding behaviour in individual allotments. Any assessment of flooding in individual allotments may also require a site survey.
LEGEND
Modelled Stormwater Network
Two-Dimensional Model Boundary
Catchment Boundary
Lyall & Associates
BLACKMANS SWAMP CREEK AND PLOUGHMANS CREEK FLOOD STUDY
Figure 6.20
(Sheet 7 of 7)
FLOOD PLANNING AREA AND LEVELS
MAIN STREAM FLOODING AND MAJOR OVERLAND FLOW AFFECTED AREAS

4A.6 ANNEXURE 2 - LAND USE CATEGORIES

4A.6.1 Essential Community Facilities

Place of Assembly or Public Buildings that may provide an important contribution to the notification and evacuation of the community during flood events. Including: Place of public worship, public administration buildings, schools, community facilities.

4A.6.2 Critical Utilities

Electricity generating works, Utility Installations that may cause pollution of waterways during flooding, are essential to evacuation during periods of flood, or if affected during flood events would unreasonably affect the ability of the community to return to normal activities after the flood events. This Includes: Air transport facilities, Emergency services facilities, Hazardous storage establishments, Hospitals, passenger transport facilities, public utility infrastructure, resource recovery facilities, sewage reticulation systems, sewage treatment plants, telecommunications facilities, water supply systems, waste disposal facilities, waste or resource transfer stations.

4A.6.3 Subdivision and Filling

Subdivision of land involving the creation of new allotments for any particular purpose (excludes boundary adjustments), earthworks or filling operations covering 100m² or more than 0.3m deep.

4A.6.4 Residential

All forms of Residential Accommodation; Tourist and Visitor Accommodation (other than forms referred to elsewhere); Residential care facilities; Group homes; Home industry; Home occupation; Housing for aged or disabled persons; Health consulting rooms; Public utility undertakings (other than critical utilities); Recreation facilities (indoor and major); and Utility installation undertaking (other than critical utilities).

4A.6.5 Commercial or Industrial

Specialised retail premises; Vehicle repair stations; Registered clubs; Commercial Premises (other than forms referred to elsewhere); Education establishment (other than forms referred to elsewhere); Retail premises; medical centres; Pubs; small bars; Industry; Institution; Junk yard; Liquid fuel depot; Hotel or Motel accommodation; Vehicle sales or hire premises; Function centres; Place of Assembly (other than essential community facilities); Place of Public worship; Public building (other than essential community facilities); Freight transport facilities; Rural industry (other than stock and sale yards); Service station; Shop; Transport terminal; Warehouse.

4A.6.6 Recreation or Agricultural

Recreation Facility (outdoor); Agriculture; Animal boarding or training establishment; Garden centre; Recreation area; Roadside stall; Stock and Sale Yard.

4A.6.7 Minor Development

a) In the case of residential development:

   i) an addition to an existing dwelling of not more than 15% or 30m² (whichever is the lesser) of the habitable floor area which existed at the date of this policy; or

   ii) the construction of an outbuilding, with walls that may impede the flood path, with a maximum floor area of 45m² - no more than 50% of the lot size can be developed; or

   iii) carports, awnings or similar structures that do not contain walls.

b) In the case of shops and offices:

   i) new shops with a total floor area less than 50m²; or

   ii) change of use which involves no building.

c) In the case of other development an addition to existing premises of not more than 15% of the floor area which existed at the date of commencement of this policy.

d) In the case of an allotment partially affected by flood liable zones new development where survey information is provided supporting the location of the development as outside the defined 1% AEP flood level.

4A.7 ANNEXURE 3 - PRESCRIPTIVE DEVELOPMENT CONTROLS – FLOOD RESPONSE LEVELS

	FLOOD PLANNING AREA								OVERLAND FLOW FLOODING						
PLANNING CONSIDERATIONS	ESSENTIAL COMMUNITY FACILITIES	CRITICAL UTILITIES	SUBDIVISION AND FILLING	RESIDENTIAL	COMMERCIAL OR INDUSTRIAL	RECREATION OR AGRICULTURE	MINOR DEVELOPMENT	PLANNING CONSIDERATIONS	ESSENTIAL COMMUNITY FACILITIES	CRITICAL UTILITIES	SUBDIVISION AND FILLING	RESIDENTIAL	COMMERCIAL OR INDUSTRIAL	RECREATION OR AGRICULTURE	MINOR DEVELOPMENT
FLOOR LEVELS				1	1		3	FLOOR LEVELS	1	2		2			
BUILDING COMPONENTS				1	1	1	1	BUILDING COMPONANTS	1	1		1			
STRUCTURAL SOUNDNESS				1,3	1,2,3	2	1,3	STRUCTURAL SOUNDNESS	1	1		2			
EVACUATION				1	1	1	1	EVACUATION	1	1	1	1	1	1	1
FLOOD AFFECTATION				1,2,5	1	2,3	1,6	FLOOD AFFECTATION	1,2,5		1,2,4	4	1,2		
MANAGEMENT AND DESIGN				2	1,2	2	2	MANAGEMENT AND DESIGN	1,2	2	2,3	2	1,2	2	2

Unsuitable Land Use

Not Relevant

Design Element	Flood Response Level	FLOOD RESPONSE CONTROLS
Floor Level	**1**	Habitable floor levels to be equal to or greater than the 1% AEP Flood level plus 0.5m freeboard
	2	Habitable floor levels to be equal to or greater than the 1% AEP Flood level plus 0.3m freeboard
	3	Floor levels to be as close to the 1% AEP Flood level as practical & no lower than any existing floor level
Building Components and Method	**1**	All structures to have flood compatible building and components below the 1% AEP Flood level
Structural Soundness	**1**	Engineers report to certify that any structure can withstand the forces of floodwater, debris & buoyancy up to and including the 1% AEP Flood level
	2	Applicant to demonstrate that any structure can withstand the forces of floodwater, debris & buoyancy up to and including the 1% AEP Flood level
	3	Geotechnical Engineers report required to specify appropriate filling/earthworks & the means of protecting batters against scouring/erosion
Evacuation	**1**	The development is to be consistent with any relevant flood evacuation strategy
Flood Affectation	**1**	Engineers report required to certify that the development will not increase flood affection elsewhere
	2	Development shall not block the conveyance of flood waters across the floodway or overland flow
	3	Filling of land up to 0.5m above 1% AEP Flood level permitted provided this does not result in any significant effect on the conveyance of flood waters
	4	Filling of a maximum of a 1/3 of allotment up to 0.3m above the 1% AEP Flood level permitted provided this does not result in any significant effect on the conveyance of flood waters or flood levels
	5	Filling of a maximum of a 1/4 of allotment up to 0.5m above the 1% AEP Flood level permitted provided this does not result in any significant effect on the conveyance of flood waters
	6	No filling
Management and Design	**1**	Applicant to demonstrate that area is available to store goods above the 1% AEP Flood level plus 0.5m freeboard
	2	No external storage of materials below the design floor level which may cause pollution or be potentially hazardous during any flood
	3	Applicant to demonstrate the at the time of Development Application lodgement that the potential development complies with this DCP as a consequence of a subdivision proposal and any potential dwelling construction can be undertaken in accordance with this Plan

4A.8 ANNEXURE 4 - FLOOD COMPATIBLE MATERIALS

Building Component	Floor Compatible Material	
Flooring and Sub-floor Structure	✓ concrete slab-on-ground monolith construction ✓ suspension reinforced concrete slab.	✓ T&G flooring and timber joists (for minor extensions)
Floor Covering	✓ clay tiles ✓ concrete, precast or in situ ✓ concrete tiles ✓ epoxy, formed-in-place ✓ mastic flooring, formed-in-place	✓ rubber sheets or tiles with chemical-set adhesives ✓ silicon floors formed-in-place ✓ vinyl sheets or tiles with chemical-set adhesive ✓ ceramic tiles, fixed with mortar or chemical-set adhesive ✓ asphalt tiles, fixed with water resistant adhesive
Wall Structure	✓ solid brickwork, blockwork, reinforced, concrete or mass concretes	✓ Gal metal and timber frame (for minor extensions)
Roofing Structure (for situations where the relevant flood level is above the ceiling)	✓ reinforced concrete construction ✓ galvanised metal construction	
Doors	✓ solid panel with water proof adhesives ✓ flush door with marine ply filled with closed cell foam	✓ painted metal construction ✓ aluminium or galvanised steel frame
Wall And Ceiling Linings	✓ fibro-cement board ✓ brick, face or glazed ✓ clay tile glazed in waterproof mortar ✓ concrete ✓ concrete block	✓ steel with waterproof applications ✓ stone, natural solid or veneer, waterproof grout ✓ glass blocks ✓ glass ✓ plastic sheeting or wall with waterproof adhesive
Insulation Windows	✓ foam (closed cell types) ✓ aluminium frame with stainless steel rollers or similar corrosion and water resistant material	✓ Gal metal and timber windows (for minor extensions)
Nails, Bolts, Hinges and Fittings	✓ Brass, nylon or stainless steel ✓ Removable pin hinges	✓ Hot dipped galvanised steel wire nails or similar

(next page is Page 5.1)